Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

1. Marginal PMF

1.1 Definition

For discrete random variables X and Y, the joint PMF is:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The marginal PMFs are obtained by summing over the other variable:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea

The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

1.2 Marginal PMF from Joint PMF Table

From a joint PMF table:

- Row sums $\rightarrow p_X(x)$
- Column sums $\rightarrow p_Y(y)$

Formally:

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

Also:

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

1.3 Example

Given joint PMF:

	$y = 0$	$y = 1$
$x = 0$	$1/8$	$1/4$
$x = 1$	$3/8$	$1/4$

Step-by-step solution
Compute $p_X(x)$:

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

Compute $p_Y(y)$:

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final Answer

$$p_X(x) = \begin{cases} 3/8 & x = 0 \\ 5/8 & x = 1 \end{cases} \quad p_Y(y) = \begin{cases} 1/2 & y = 0 \\ 1/2 & y = 1 \end{cases}$$

2. Correlation

2.1 Definition

$$R_{XY} = E[XY]$$

It represents the average value of the product XY.

2.2 Interpretation and Limitation

- R_{XY} depends on the scale of X and Y
- Computed about the origin, not about means
- Mixes mean values, spread, and dependence

Important Result

$$R_{XY} = 0 \Rightarrow \text{orthogonal about the origin}$$

Transition
To remove mean effects $\rightarrow$ center variables $\rightarrow$ covariance

3. Covariance

3.1 Definition

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Measures how centered variables vary together.

3.2 Interpretation Summary

- $\mathrm{Cov}(X, Y) > 0 :$ positive covariance
- $\mathrm{Cov}(X, Y) < 0 :$ negative covariance
- $\mathrm{Cov}(X, Y) = 0 :$ no linear co-variation

Indicates direction only; magnitude depends on units.

3.3 Algebraic Form (Derivation)

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

Step 1: Expand

$$= E[XY - X\mu_Y - \mu_X Y + \mu_X\mu_Y]$$

Step 2: Apply linearity

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Step 3: Substitute means

$$= E[XY] - \mu_X\mu_Y$$

Final Result

$$\boxed{\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]}$$

3.4 Properties

1. $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$
2. $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$
3. $\mathrm{Cov}(aX+b, cY+d) = ac \cdot \mathrm{Cov}(X,Y)$
4. $X \perp Y \Rightarrow \mathrm{Cov}(X,Y) = 0$

Zero covariance does NOT imply independence.

3.5 Example 1

Given:

$$Y = -2X + 3$$

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

$$XY = -2X^2 + 3X$$

$$E[XY] = -2E[X^2] + 3E[X]$$

$$E[Y] = -2E[X] + 3$$

$$\mathrm{Cov}(X,Y) = -2E[X^2] + 2(E[X])^2$$

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

$$\Rightarrow \mathrm{Cov}(X,Y) = -2\mathrm{Var}(X)$$

4. Pearson's Correlation Coefficient

4.1 Definition

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

4.2 Properties

$$-1 \leq \rho_{XY} \leq 1$$

Measures strength and direction of linear relationship.

4.3 Interpretation

- $\rho_{XY} > 0:$ positive trend
- $\rho_{XY} < 0:$ negative trend
- $\rho_{XY} = 0:$ no linear correlation

5. Example 2 (Continuous Case)

$$f_{XY}(x,y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4,\ 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \frac{8}{3}, \quad E[Y] = \frac{31}{9}$$

$$E[XY] = \frac{248}{27}$$

$$E[2X + 3Y] = \frac{47}{3}$$

$$\mathrm{Var}(X) = \frac{8}{9}, \quad \mathrm{Var}(Y) = \frac{92}{81}$$

$$\mathrm{Cov}(X,Y) = 0, \quad \rho_{XY} = 0$$

6. Jointly Gaussian Random Variables

6.1 Definition

A pair (X, Y) is jointly Gaussian if:

- Joint density is bivariate Gaussian
- Marginals are Gaussian

6.3 Joint Gaussian PDF

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y\right.\right.\right.$$

Final Key Takeaways

1. Marginal PMF = sum over joint PMF
2. Correlation is raw and not centered
3. Covariance removes mean effect
4. $\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$
5. Zero covariance $\neq$ independence
6. Pearson coefficient normalizes covariance
7. Joint Gaussian fully described by means, variances, and correlation